

Answer the problems on **separate** paper. You do not need to rewrite the problem statements on your answer sheets. Do your own work. Show **all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (32 pts) For each of the following functions find  $\frac{dy}{dx}$ :

a.  $y = \sqrt{\frac{x-1}{x^2+1}}$

b.  $y = \sec^2(x - x^2)$

c.  $y = x^3 \sqrt{x^2 - x - 2}$

d.  $y = \frac{x^2 - 2}{\cos 3x}$

2. (8 pts) Find, using implicit differentiation,  $\frac{dy}{dx}$ :

a.  $x^2 - 4xy^3 + y^4 = x - 2y$

3. (8 pts) Find the equation of the tangent line to the curve  $y = (2x^2 - x - 3)^3$  at  $x = 1$ .

4. (24 pts) Evaluate the integrals:

a.  $\int \sqrt{4 - 3x^2} x \, dx$

b.  $\int (2 + 3x^2)^2 \, dx$

c.  $\int x^2 \cos(4x) \, dx$

5. (8 pts) Find the partial fraction decomposition of:

a.  $\frac{2x-1}{x(x+3)^2}$

6. (16 pts) Let  $\alpha = 3 - 2i$ ,  $\beta = 2 + 5i$ ,  $\gamma = -2 + 4i$ . Find (in rectangular form):

a.  $\alpha^2 + 2i\bar{\beta}$

b.  $\frac{\alpha}{\beta - 2\gamma}$

7. (8 pts) Find all of the square roots of  $\alpha = \sqrt{3} - i$ .